Cross-Generator Detection of AI-Generated Product Reviews: A Leave-One-Generator-Out Study with Contrastive and Adversarial Robustness Objectives

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Abstract

Detectors for machine-generated text are usually trained and evaluated on output from the same language model, which inflates their apparent reliability. We study the harder and more realistic question of *cross-generator* robustness: does a human-versus-AI review detector still work on reviews from a generator it never saw during training? We build a balanced corpus of roughly 20,000 Amazon product reviews, pairing about 10,000 confidently human reviews (written before the public release of ChatGPT) with about 10,000 AI reviews evenly drawn from four distinct generators (GPT-5 mini, DeepSeek-V4-Flash, Gemma-4-31B-it, and Qwen3.6-35B-A3B). All AI reviews describe the same products with lengths sampled from the human length distribution, so that style rather than length or topic is the variable under test. We adopt a leave-one-generator-out (LOGO) protocol with four rotations, each holding out one generator entirely as a cross-generator test set. On top of a DeBERTa-v3 encoder we train four variants that toggle a supervised contrastive head and a gradient-reversal generator-adversarial head, and we compare them against TF-IDF and frozen-DeBERTa logistic-regression baselines. Fine-tuned DeBERTa generalizes strongly across generators (cross-generator F1 of 0.9839 for plain cross-entropy versus 0.9558 for TF-IDF), and the supervised contrastive objective both raises cross-generator F1 to 0.9875 and roughly halves the in-distribution-to-cross-generator gap from 0.0095 to 0.0048. The adversarial objective produces the smallest gap (0.0015) but at a cost to absolute accuracy. We conclude that representation-shaping objectives measurably improve cross-generator robustness, while noting that the controlled, non-adversarial generation setting likely makes detection easier than in the wild.

1 Introduction

Large language models can now produce fluent, on-topic product reviews that are difficult to distinguish from human writing. This creates an obvious incentive for review fraud, and a matching need for automatic detectors of AI-generated reviews. A detector is only useful, however, if it works on text from models it was not trained on. New generators appear constantly, and an attacker is free to use whichever model the defender has not yet seen. Most prior detection work trains and tests on a single generator, or pools several generators into both train and test, which measures memorization of a fixed set of model fingerprints rather than generalization to a new one. The resulting accuracy numbers tend to overstate real-world performance.

This project measures the quantity that actually matters for deployment: the *cross-generator generalization gap*, the drop in detection F1 when a detector is evaluated on a generator absent from its training data. We frame the problem as binary classification (human versus AI) over

Amazon product reviews and evaluate under a leave-one-generator-out (LOGO) protocol. With four generators we run four rotations; in each, one generator is held out entirely as the cross-generator test set and the detector is trained on human reviews plus the other three generators. The headline number is the difference between F1 on an in-distribution test set (generators seen in training) and F1 on the held-out generator.

We make three contributions. First, we construct a controlled corpus in which length and topic are held fixed across generators, so the cross-generator gap reflects writing *style* rather than a length or category shortcut. Second, we implement and compare four detector variants on a shared DeBERTa-v3 encoder: plain cross-entropy, a supervised contrastive variant, a generator-adversarial variant using a gradient-reversal layer, and the combination. Third, we report a full LOGO evaluation against two classical baselines and analyze which generators are hardest to catch unseen. Our central finding is that a fine-tuned transformer already generalizes well in this setting, and that a supervised contrastive objective improves cross-generator F1 while approximately halving the generalization gap.

2 Related Work

Machine-generated text detection. Detecting text produced by language models has been approached with supervised classifiers fine-tuned on labeled human/AI corpora, with zero-shot statistical methods that score a passage under a reference model's likelihood, and with watermarking schemes applied at generation time. Supervised detectors are accurate in-distribution but are known to degrade when the test text comes from a different model, decoding strategy, or domain than the training data. Our work isolates one axis of that degradation, the change of generator, and measures it directly rather than assuming a fixed train and test generator.

Fake and AI-generated review detection. A line of recent work studies AI-generated reviews specifically, motivated by the threat to online marketplaces. These studies report strong same-generator accuracy but generally do not separate the generator used for training from the one used for testing. Our LOGO design is a deliberate departure: it never lets a generator appear in both training and the cross-generator test split, so the reported gap is a clean estimate of generalization to an unseen model.

Robust and transferable representations. Two ideas from representation learning motivate our model variants. Supervised contrastive learning [3] pulls together examples sharing a label and pushes apart examples with different labels, which in our setting should cluster all AI reviews together and all human reviews together regardless of which model wrote the AI text, encouraging a generator-invariant notion of "what AI writing looks like." Domain-adversarial training with a gradient reversal layer [4] trains an encoder so that an auxiliary head *cannot* recover the domain (here, the source generator), removing generator-specific cues from the representation. We treat the source generator as the adversarial domain label. Our encoder backbone is DeBERTa-v3 [2], a disentangled-attention transformer that is strong on sentence- and document-level classification.

Table 1: Final corpus composition after cleaning.

Source	Count
Human reviews	9,969
GPT-5 mini	2,490
DeepSeek-V4-Flash	2,499
Gemma-4-31B-it	2,500
Qwen3.6-35B-A3B	2,498
Total	**19,956**

3 Methodology

3.1 Data sources

The corpus combines two sources into a single labeled set of human versus AI-generated Amazon product reviews. Human reviews are sampled from the McAuley-Lab Amazon Reviews 2023 dataset [1], restricted to reviews written before ChatGPT's public release. AI reviews are generated by us, prompting four different language models to write customer-style reviews for products drawn from the human pool. Table 1 gives the final counts after cleaning.

3.2 Human review collection

The Amazon Reviews 2023 per-category files are multi-gigabyte, so we stream them line by line and draw a uniform sample with reservoir sampling (Algorithm R). Each candidate review must satisfy three filters: a pre-ChatGPT timestamp (before 2022-11-30 00:00:00 UTC), giving high confidence the text is human-written; a length between 20 and 400 words, excluding trivially short and outlier-long reviews; and an English-only language check via `langdetect`, since the dataset contains a small fraction of non-English reviews. We sample 1,250 reviews from each of eight categories (All_Beauty, Books, Electronics, Home_and_Kitchen, Sports_and_Outdoors, Toys_and_Games, Pet_Supplies, Office_Products) for a 10,000 target. The resulting human word counts have a mean of 60.6 and a median of 43, and the rating distribution is heavily skewed toward 5 stars, which we propagate into the AI prompts.

3.3 AI review generation

The central question requires multiple generators with varied training data, architectures, and providers, so we chose four deliberately distinct families: `gpt5mini` (`gpt-5-mini`, OpenAI API), `deepseek` (`DeepSeek-V4-Flash`), `gemma` (`gemma-4-31B-it`), and `qwen` (`Qwen3.6-35B-A3B`), the latter three served through the HuggingFace Inference Providers router. The original proposal listed IBM Granite as the fourth generator; we substituted DeepSeek-V4-Flash because no Granite model is served through commodity hosted APIs, while preserving the property we wanted, a distinct open-weight family.

For each generator we sample the same 2,500 unique products from the human pool (identical random seed across generators), so every generator describes the same product set. The prompt template is identical across generators and is parameterized by the product title, category, target rating, and a target word count. Critically, the target word count for each AI review is drawn from the empirical distribution of human review word counts. If AI reviews were systematically longer

Table 2: The four leave-one-generator-out rotations.

Rotation	Held out (cross-gen)	Training generators
0	gpt5mini	deepseek + gemma + qwen
1	deepseek	gpt5mini + gemma + qwen
2	gemma	gpt5mini + deepseek + qwen
3	qwen	gpt5mini + deepseek + gemma

or shorter than human ones, a detector could exploit length alone and the cross-generator results would say nothing about style. The prompt also bans emojis and hashtags and asks for the review body only. Generation uses `temperature` = 1.0, `top_p` = 0.95, and a generous `max_tokens` cap that accommodates the internal reasoning tokens that three of the four generators emit. Per-generator output word counts (means of 59.9 to 68.6) closely track the human mean of 60.6 by construction.

3.4 Data cleaning

Two issues required post-hoc cleaning. A `langdetect` sweep removed 31 non-English human reviews and 11 non-English AI reviews. Reasoning-model contamination required more care: 289 Qwen reviews were pure internal reasoning traces and were deleted (disabling thinking server-side via `chat_template_kwargs` fixed the remainder); 97 DeepSeek reviews leaked `<think>` blocks and were salvaged by stripping everything up to the closing tag, and 19 further DeepSeek reviews truncated mid-reasoning were dropped. These cleanups left non-round per-generator counts, which the split builder is designed to absorb.

3.5 Leave-one-generator-out splits

To measure cross-generator robustness directly, we use four LOGO rotations (Table 2). In each rotation, one generator is held out entirely as the cross-generator test set, and the model trains on human reviews plus the other three generators.

Each rotation produces four splits. The three training generators' reviews are split 80/10/10 into `train`, `val_indist`, and `test_indist`; the held-out generator's entire set plus a disjoint human pool forms `test_crossgen`. We carve the human pool *after* the AI split, taking exactly as many humans as the AI in-distribution count so that each in-distribution split is exactly 1:1 balanced, and assigning all remaining humans (about 2,470 per rotation) to the cross-generator pool. Because the cross-generator humans are disjoint from the training humans, cross-generator evaluation measures generalization to an unseen generator rather than memorization of human text; the split builder asserts no `review_id` overlap between `train` and `test_crossgen`. This adaptive carving removes the original code's assumption of exactly 10,000 humans and 2,500 reviews per generator. Table 3 reports the resulting sizes.

3.6 Model and objectives

The detector is a DeBERTa-v3-base encoder [2] with up to three heads on the mean-pooled review representation h. The classification head is a linear layer trained with cross-entropy on the human/AI label; this head *is* the detector, and the other two heads exist only to shape the encoder. The contrastive projection head is a linear layer to 128 dimensions followed by L2 normalization, trained with supervised contrastive loss [3] using the binary label, which pulls AI together and

Table 3: Split sizes per rotation. In-distribution splits are exactly 1:1 human/AI; the cross-generator split is near-1:1.

Rotation	train	val_indist	test_indist	test_crossgen (AI + human)
0 (gpt5mini out)	11,994	1,496	1,504	4,962 (2,490 + 2,472)
1 (deepseek out)	11,980	1,496	1,500	4,980 (2,499 + 2,481)
2 (gemma out)	11,978	1,494	1,502	4,982 (2,500 + 2,482)
3 (qwen out)	11,982	1,496	1,500	4,978 (2,498 + 2,480)

Table 4: The four DeBERTa variants and their active loss terms.

Variant	Active loss
`base` (CE)	$\mathcal{L}_{\text{ce}}$
`contrastive`	$\mathcal{L}_{\text{ce}} + \lambda_c \mathcal{L}_{\text{supcon}}$
`adversarial`	$\mathcal{L}_{\text{ce}} + \lambda_a \mathcal{L}_{\text{adv}}$
`both`	$\mathcal{L}_{\text{ce}} + \lambda_c \mathcal{L}_{\text{supcon}} + \lambda_a \mathcal{L}_{\text{adv}}$

human together regardless of generator. The generator-adversarial head is a linear layer predicting which source wrote the review (the three training generators plus `human`, four classes), fed through a gradient reversal layer [4] so that minimizing its loss *maximizes* generator confusion in the encoder, forcing it to drop generator-specific style cues. The total loss is

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \lambda_c \, \mathcal{L}_{\text{supcon}} + \lambda_a \, \mathcal{L}_{\text{adv}}, \tag{1}$$

with $\lambda_c = \lambda_a = 0.5$. The gradient reversal layer's strength λ_a is ramped from 0 upward over training in DANN style, since adversarial training is unstable at full strength from the first step. The four variants simply toggle which terms are active, as shown in Table 4.

3.7 Baselines

We compare against two classical baselines, each run per rotation. The first is TF-IDF features with logistic regression fit on the training text. The second is a frozen DeBERTa-v3 encoder (no fine-tuning) with logistic regression on the pooled features, a linear probe that shows what fine-tuning buys over raw pretrained features.

3.8 Training configuration

All variants share the configuration in Table 5. We select the best epoch on `val_indist` F1, fix all seeds at 42, and run each of the four rotations independently. Float32 precision is used to avoid NaNs observed on the local RTX 4060 GPU.

4 Experiments and Results

We report F1 with the positive class being AI, and ROC-AUC, on both the in-distribution test set and the held-out cross-generator set, averaged across the four LOGO rotations. The headline metric is the gap, in-distribution F1 minus cross-generator F1.

Table 5: Training configuration, shared across all variants and rotations.

Setting	Value
Learning rate	2e-5
Epochs	3
Batch size	4 (effective 16 with gradient accumulation 4)
Max sequence length	256
Precision	float32 (avoids NaNs on 4060)
Optimizer	AdamW, weight decay 0.01, 100 warmup steps
Seed	42

Table 6: Baseline LOGO results, averaged over four rotations.

Baseline	In-Dist F1	In-Dist AUC	Cross-Gen F1	Gap (F1)
TF-IDF + LogReg	0.9709	0.9964	0.9558	0.0151
Frozen DeBERTa + LogReg	0.9657	0.9950	0.9437	0.0220

4.1 Baselines

Table 6 reports the two baselines. TF-IDF + logistic regression is a remarkably strong in-distribution detector (0.9709 F1) but loses about 1.5 F1 points cross-generator (0.9558, gap 0.0151). The frozen-DeBERTa linear probe is weaker overall (0.9437 cross-generator F1) and less stable across rotations, with a large 0.0589 gap when DeepSeek is held out. Both baselines confirm that shallow lexical and frozen-feature detectors carry a real cross-generator penalty.

4.2 DeBERTa variants

Table 7 is the central result. Fine-tuning DeBERTa lifts cross-generator F1 from the 0.94 to 0.96 baseline range to 0.9831 or above for every variant, and pushes cross-generator AUC above 0.995. Among the variants, the supervised contrastive objective is the best overall detector: it achieves the highest cross-generator F1 (0.9875) and the highest cross-generator AUC (0.9994), while halving the gap relative to plain cross-entropy (0.0048 versus 0.0095). The adversarial objective produces the smallest gap of all (0.0015) but lowers both in-distribution and cross-generator F1, indicating it trades absolute accuracy for invariance. Combining both objectives does not beat contrastive alone here; the adversarial term appears to slightly destabilize the contrastive clustering. Figures 1 and 2 visualize these comparisons.

4.3 Per-held-out-generator breakdown

Table 8 and Figure 3 break cross-generator F1 down by which generator was held out. Gemma (rotation 2) is consistently the hardest generator to catch unseen: plain cross-entropy drops to 0.9681 on held-out Gemma, its worst rotation. Both representation-shaping objectives help most exactly here, with contrastive raising held-out-Gemma F1 to 0.9808 and adversarial to 0.9815. GPT-5 mini and Qwen are the easiest to detect unseen, with all variants above 0.98. This pattern explains why contrastive learning improves the average: its largest gains land on the rotation where the base model is weakest.

Table 7: Main comparison: DeBERTa variants, averaged over four LOGO rotations. Best value in each column is in bold.

Model	In-Dist F1	Cross-Gen F1	Gap (F1)	Cross-Gen AUC
DeBERTa + CE (base)	**0.9933**	0.9839	0.0095	0.9991
+ Contrastive	0.9924	**0.9875**	0.0048	**0.9994**
+ Adversarial	0.9847	0.9832	**0.0015**	0.9964
+ Both	0.9901	0.9831	0.0069	0.9954

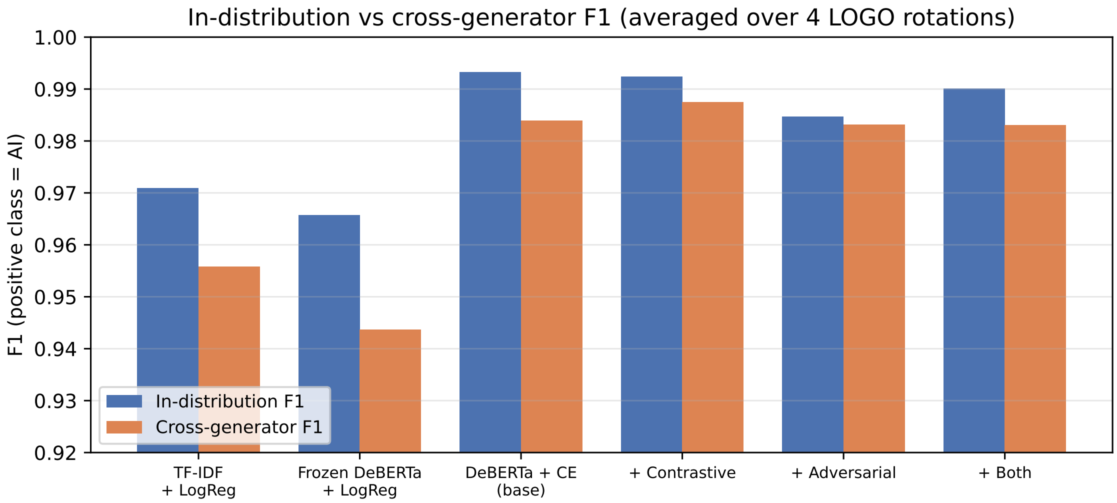

Figure 1: In-distribution versus cross-generator F1 for the two baselines and the four DeBERTa variants, averaged over the four LOGO rotations. Fine-tuned DeBERTa closes most of the cross-generator gap that the baselines suffer.

5 Discussion and Analysis

Fine-tuning is what closes most of the gap. The single largest improvement in cross-generator F1 comes not from the robustness objectives but from fine-tuning the encoder at all: cross-generator F1 rises from 0.9437 (frozen probe) and 0.9558 (TF-IDF) to 0.9839 for plain cross-entropy. A fine-tuned DeBERTa already learns a fairly generator-invariant notion of AI writing in this corpus. The robustness objectives then provide a second, smaller improvement on top.

Contrastive learning gives the best practical detector. The supervised contrastive variant is the one we would deploy. It has the highest cross-generator F1 and AUC of any model, and it halves the generalization gap relative to plain cross-entropy without the accuracy cost the adversarial variant pays. Pulling all AI reviews into one cluster and all human reviews into another, regardless of source, is exactly the inductive bias the cross-generator task rewards, and its biggest gains appear on the hardest rotation (held-out Gemma).

Adversarial training minimizes the gap but not the error. The gradient-reversal variant produces the smallest gap (0.0015) precisely because it lowers in-distribution F1 toward the cross-generator level rather than raising cross-generator F1. Minimizing the gap is therefore not the same

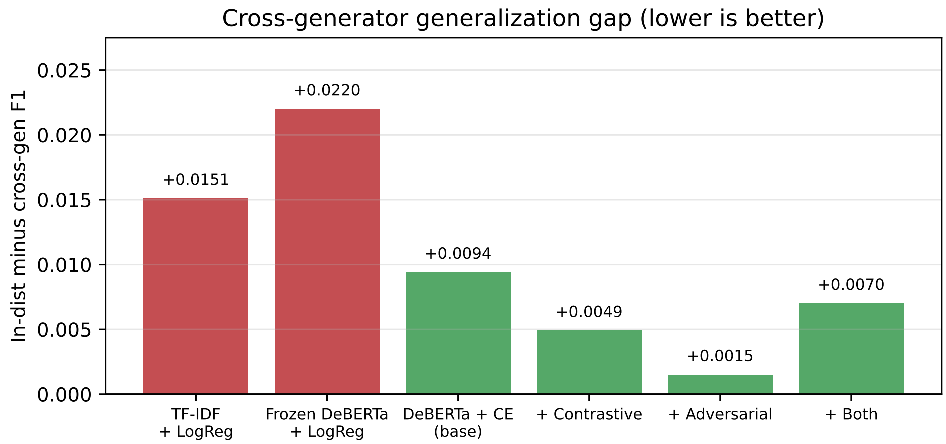

Figure 2: Cross-generator generalization gap (in-distribution F1 minus cross-generator F1) by model. The contrastive and adversarial objectives both shrink the gap relative to plain cross-entropy.

Table 8: Cross-generator F1 by held-out generator (one rotation each). Best value in each row is in bold.

Rotation	Held out	CE	Contrastive	Adversarial	Both
0	gpt5mini	0.9912	**0.9902**	0.9875	0.9844
1	deepseek	0.9843	**0.9861**	0.9831	0.9802
2	gemma	0.9681	0.9808	**0.9815**	0.9754
3	qwen	0.9918	**0.9930**	0.9807	0.9926

as maximizing detection quality; a model that is uniformly mediocre has a small gap. This is worth stating because the gap was our proposed headline metric: it should be read together with absolute F1, not alone. Combining the two objectives did not help, suggesting the adversarial gradient and the contrastive clustering work against each other at these loss weights.

What the controlled setting does and does not show. Because target lengths were sampled from the human distribution and all generators wrote about the same products, our gap reflects writing style, not length or topic. That is the right design for the scientific question, but it also makes detection easier than in the wild, where fake reviews often are systematically shorter or longer and cluster on particular products. Our absolute F1 numbers should be read as an upper bound on a controlled setting, not as field performance.

Limitations. Several caveats apply. We substituted DeepSeek-V4-Flash for the proposal's IBM Granite because Granite is not served through commodity hosted APIs, so our results speak to the GPT-5 / DeepSeek / Gemma / Qwen family specifically. The corpus is English-only. Three of four generators are reasoning models; although we removed obvious reasoning traces, subtler stylistic artifacts of reasoning models could remain and act as a shortcut feature. A single fixed prompt template was used, whereas a real adversary would tune prompts, so our setting measures detection of controlled, non-adversarial generation. Finally, each rotation's cross-generator human pool is

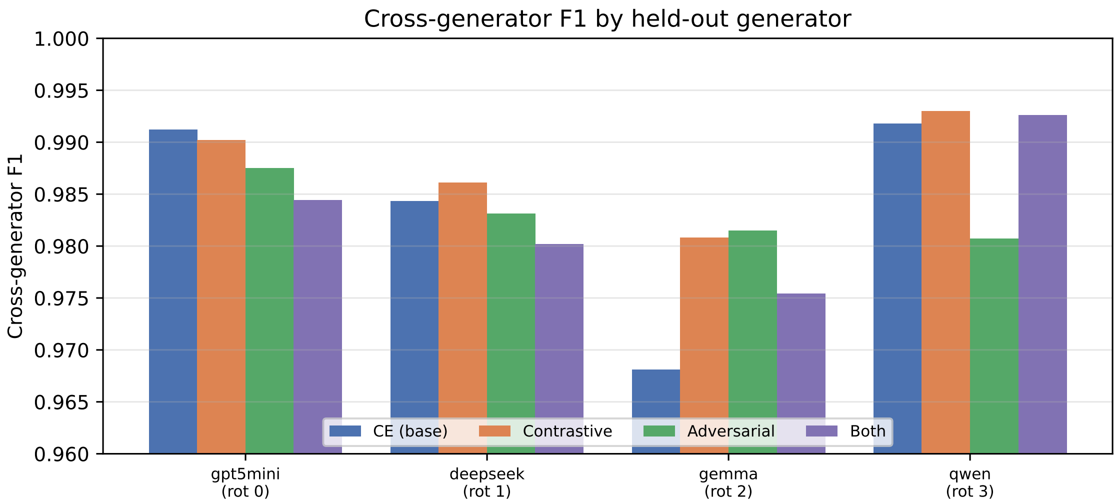

Figure 3: Cross-generator F1 by held-out generator for the four DeBERTa variants. Gemma (rotation 2) is the hardest to detect unseen, and it is where the contrastive and adversarial objectives help most.

modest (about 2,470), which adds sampling noise to the per-rotation cross-generator estimates, though it remains adequate for F1 and AUC.

6 Conclusion

We studied whether an AI-review detector generalizes to generators it never saw in training, using a controlled corpus of about 20,000 Amazon reviews and a strict leave-one-generator-out protocol that holds product, topic, and length fixed so the test isolates writing style. Fine-tuned DeBERTa-v3 generalizes far better across generators than TF-IDF or frozen-feature baselines, and a supervised contrastive objective is the best variant overall: it achieves the highest cross-generator F1 (0.9875) and AUC (0.9994) and halves the generalization gap relative to plain cross-entropy (0.0048 versus 0.0095). A gradient-reversal adversarial objective minimizes the gap further (0.0015) but by lowering in-distribution accuracy rather than raising cross-generator accuracy, which shows that the gap must be read alongside absolute error. The hardest generator to catch unseen is Gemma, and that is exactly where the representation-shaping objectives help most. Future work should test against genuinely held-out model families, prompt-tuned adversarial generation, and non-English reviews, and should explore whether the contrastive and adversarial objectives can be combined without the interference we observed.

Contribution Statement

Placeholder: describe each team member's contributions to data collection, generation, split construction, model implementation, training, evaluation, and the writing of this report.

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